Practical 14

Sudo mysql

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CREATE DATABASE Bank14;
USE Bank14;

CREATE TABLE Customer (
    cust_id INT PRIMARY KEY,
    cust_name VARCHAR(50) NOT NULL,
    city VARCHAR(50) NOT NULL,
    phone VARCHAR(15)
);

CREATE TABLE Branch (
    branch_id INT PRIMARY KEY,
    branch_name VARCHAR(50) NOT NULL,
    branch_city VARCHAR(50) NOT NULL
);

CREATE TABLE Account (
    acc_no INT PRIMARY KEY,
    cust_id INT NOT NULL,
    branch_id INT NOT NULL,
    balance DECIMAL(12,2) NOT NULL,
    FOREIGN KEY (cust_id) REFERENCES Customer(cust_id),
    FOREIGN KEY (branch_id) REFERENCES Branch(branch_id)
);

CREATE TABLE Loan (
    loan_no INT PRIMARY KEY,
    cust_id INT NOT NULL,
    branch_id INT NOT NULL,
    amount DECIMAL(12,2) NOT NULL,
    FOREIGN KEY (cust_id) REFERENCES Customer(cust_id),
    FOREIGN KEY (branch_id) REFERENCES Branch(branch_id)
);

INSERT INTO Customer VALUES
(1, 'Aarav', 'Pune', '9876543210'),
(2, 'Priya', 'Mumbai', '9876543211'),
(3, 'Pooja', 'Pune', '9876543212'),
(4, 'Rahul', 'Nashik', NULL),
(5, 'Pranav', 'Mumbai', '9876543214'),
(6, 'Sneha', 'Pune', NULL),
(7, 'Vikram', 'Delhi', '9876543216'),
(8, 'Payal', 'Mumbai', '9876543217'),
(9, 'Neha', 'Nagpur', NULL),
(10, 'Karan', 'Pune', '9876543219');

INSERT INTO Branch VALUES
(101, 'FC Road Branch', 'Pune'),
(102, 'Andheri Branch', 'Mumbai'),
(103, 'Nashik Road Branch', 'Nashik'),
(104, 'Camp Branch', 'Pune'),
(105, 'Bandra Branch', 'Mumbai'),
(106, 'Nagpur Main Branch', 'Nagpur'),
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(107, 'Delhi Central Branch', 'Delhi'),
(108, 'Thane Branch', 'Thane'),
(109, 'Kothrud Branch', 'Pune'),
(110, 'Vashi Branch', 'Navi Mumbai');

INSERT INTO Account VALUES
(2001, 1, 101, 25000),
(2002, 2, 102, 40000),
(2003, 3, 104, 35000),
(2004, 4, 103, 50000),
(2005, 5, 105, 30000),
(2006, 6, 109, 45000),
(2007, 7, 107, 60000),
(2008, 8, 110, 55000),
(2009, 9, 106, 20000),
(2010, 10, 101, 65000);

INSERT INTO Loan VALUES
(501, 1, 101, 150000),
(502, 2, 102, 80000),
(503, 3, 104, 120000),
(504, 4, 103, 90000),
(505, 5, 105, 200000),
(506, 6, 109, 110000),
(507, 7, 107, 75000),
(508, 8, 110, 130000),
(509, 1, 101, 100000),
(510, 10, 101, 60000);

1. select * from Customer where city in ('Pune','Mumbai') order by c
ust_name asc;

2. select * from Customer where cust_name like '%a%';

3. select * from Customer where phone is not null;

4. select * from Customer where (city='Pune' and cust_name like 'p%')or
city='Mumbai';

5. SELECT B.branch_name, MAX(L.amount) AS max_loan, AVG(L.amount) AS avg_loan
FROM Branch B INNER JOIN Loan L ON B.branch_id = L.branch_id GROUP BY
B.branch_id, B.branch_name;

6. select b.branch_name ,sum(l.amount) as total_loan from Branch b i
nner join Loan l on b.branch_id=l.branch_id group by b.branch_id,b.branc
h_name having sum(l.amount) >100000;

7. select acc_no,balance,balance+500 as updated_balance from Account;

8. update Account set branch_id=102 where acc_no=2005;

10. a. select cust_id from Account intersect select cust_id from Loan;
b. select cust_id from Account except select cust_id from Loan;
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